

Natrx Assess BOP Site Selection

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Change Log

Revision #	Date	Description of Changes
0	Dec 17, 2025	Initial Release for Phase I deliverable.
1	Jan 29, 2026	- Correct oyster species name from "CrOssostrea" to "Crassostrea" virginica in section title. - Updated link for B7. Selected BlueTopo over Digital Coast (B1); actual dataset vs portal. - Minor edits to Executive Summary for accuracy. - Added J3 to Table 3 to find adequate park parcel information for BOP sites in Hudson River - Added M5 & M6 to Table 3 to include MS4 data for upper Hudson River sites - Added "Data Review" section. - Updated section numbering.
2	Mar 12, 2026	- Added A7 to the Data Review section. - Highlighted H2, H4 to include in the Data Review section. - Added Section 5 - Rewrote Section 2 as a concise introduction to site evaluation variables. Expanded scope to include physical and built environment categories alongside water quality. Detailed ecological rationale moved to Section 5. Updated section title.
3	May 5, 2026	- Updated section 5 to reflect methodology used. - Added Sections 6 and 7. - Added relevant references.

1. Executive Summary

Billion Oyster Project (BOP) has identified a need for systematic site evaluation and prioritization within its restoration pipeline. BOP seeks to evaluate 78 potential sites. BOP has contracted Natrx to assist with identifying 10-15 project sites with the intent to be ready for execution by 2029-2030, contingent on expected regulatory streamlining with New York State Department of Environmental Conservation (DEC). Working primarily on or in close proximity to publicly owned lands through New York Parks partnerships for broader public benefit, BOP designs subtidal and intertidal reef projects ranging from 5 to 60 acres using disparate data sources covering physical parameters, water quality metrics, and built environment constraints.

BOP's established site selection methodology, involving extensive hydrographic surveys, ecological sampling, and sediment characterization, delivers high-confidence site characterization but requires substantial resources per location. To efficiently evaluate the 78 candidate sites needed to achieve programmatic goals, BOP requires a preliminary screening framework that leverages existing spatial data to identify the most promising candidates warranting detailed field investigation. This document provides data inventory assessment for key variables, methodology development, custom data layer creation, and documentation to establish the foundation for systematic site prioritization, enabling BOP to transition from its current approach to an efficient, data-driven site selection process that significantly advances progress towards its goal of deploying one billion oyster spat across New York Harbor while maximizing restoration success rates.

2. Site Evaluation Variables

The objective of this prioritization analysis was to identify which candidate sites offer the most favorable combination of ecological conditions and practical feasibility for restoration, considering both the habitat requirements of *Crassostrea virginica* and constraints related to constructability and permitting that determine project success. The variables considered in this assessment were identified by BOP through an internal technical review documented in the **BOP Pipeline Site Selection Variables Memo** (Natrx, 2024). These variables form the basis of the scoring framework developed in Phase II and described in *Section 5* of this document.

C. virginica is a suspension feeder whose growth and survival are influenced by food availability, water clarity, oxygen levels, and salinity, the primary axes along which habitat quality varies across New York Harbor and its tributary water bodies. The water quality variables retained for evaluation were:

- Chlorophyll-a (Chl-a);
- Salinity;
- Turbidity or total suspended solids (TSS);
- Dissolved oxygen (DO)

Beyond water quality, physical parameters including depth, wave exposure, and shoreline change directly influence design requirements and long-term reef viability. Built environment constraints, including proximity to CSO and MS4 outfalls, overlap with maintained navigational channels, and access to NYC park lands also affect permitting risk and site suitability. The biological rationale and response thresholds associated with each variable are documented in *Section 5*.

3. Data Inventory

Data inventories (**Phase I**) were researched to capture the spatial and temporal variability required to understand environmental drivers for *C. virginica* as defined above. The data presented below integrates a broad suite of physical and water quality datasets, ranging from localized measurement programs to regional monitoring networks and aggregated products that may incorporate earlier sources. Data highlighted in **Tables 1-3** represent the curated subset retained to ensure relevance, consistency, and non-duplicative use

3.1 Physical

Table 1. Data Inventory - Physical parameters.

Variables	Ref ID	Data Sources	Data Type	Type
Water/Current Velocity/ Streamflow	A1	HRECOS (Hudson River Environmental Conditions Observing System)	Monitored	Point Time Series
	A2	NOAA Tides & Currents (NY/NJ Harbor PORTS)	Monitored	Point Time Series
	A3	USGS National Water Information System (NWIS) ¹	Monitored	Point Time Series
	A4	NY State Department of Environmental Conservation, Division of Water Monitoring Data Portal	Monitored	Point Time Series
	A5	NYC DEP Harbor Survey	Monitored	Point Time Series

¹ Data source will be decommissioned in favor of a newer API from the USGS.

	A6	NOAA National Data Buoy Center	Monitored	Point Time Series
	A7	New York/New Jersey Operational Forecast System (NYOFS)	Modeled	Raster
	A8	New York StreamStats	Modeled	Point Time Series
	A9	Urban Ocean Observatory at Davidson Laboratory	Modeled	Raster
	A10	Water Data (USGS)	Monitored	Point Time Series
Depth	B1	Digital Coast (NOAA)	Monitored	Raster
	B2	Water Levels (NOAA)	Monitored	Point Time Series
	B3	NYC DEP Harbor Survey	Monitored	Point Time Series
	B4	NOAA ENC (Electronic Navigational Charts)	Monitored	Polygon
	B5	USACE Hydrographic Surveys	Monitored	Mixed
	B6	Hudson River National Estuarine Research Reserve	Monitored	Point Time Series
	B7	BlueTopo (NOAA)	Modeled	Raster
	B8	FisherMap	Mapped	Map
	B9	Water Data (USGS)	Monitored	Point Time Series
Wave exposure/fetch	C1	Natrx Assess²	Modeled	Point
	C2	National Data Buoy Center (NOAA)	Monitored	Point
Shoreline Change Analysis (SCA)/Shoreline Erosion	D1	Natrx Assess³	Derived	Point

² See Appendix B for more details on wave modeling methodology.

³ See Appendix A for more details on SCA methodology.

3.2 Water Quality

Table 2. Data Inventory - Water Quality Parameters.

Variables	Ref ID	Data Sources	Data Type	Type
Chl-a	E1	NY State Department of Environmental Conservation, Division of Water Monitoring Data Portal	Monitored	Point Time Series
	E2	National Water Quality Monitoring Council	Monitored	Point Time Series
	E3	NYC DEP Harbor Survey	Monitored	Point Time Series
	E4	HRECOS (Hudson River Environmental Conditions Observing System)	Monitored	Point Time Series
	E5	Hudson River National Estuarine Research Reserve	Monitored	Point Time Series
	E6	Riverkeeper (Citizen Science)	Monitored	Point Time Series
	E7	NOAA coastwatch	Modeled	Raster
Salinity	F1	NY State Department of Environmental Conservation, Division of Water Monitoring Data Portal	Monitored	Point Time Series
	F2	National Water Quality Monitoring Council	Monitored	Point Time Series
	F3	NYC DEP Harbor Survey	Monitored	Point Time Series
	F4	HRECOS (Hudson River Environmental Conditions Observing System)	Monitored	Point Time Series
	F5	Hudson River National Estuarine Research Reserve	Monitored	Point Time Series
	F6	Harbor SEALs (Citizen Science)	Monitored	Point Time Series
Turbidity or TSS	G1	NY State Department of Environmental Conservation, Division of Water Monitoring Data Portal	Monitored	Point Time Series
	G2	National Water Quality Monitoring Council	Monitored	Point Time Series

	G3	NYC DEP Harbor Survey	Monitored	Point Time Series
	G4	Hudson River National Estuarine Research Reserve	Monitored	Point Time Series
	G5	HRECOS (Hudson River Environmental Conditions Observing System)	Monitored	Point Time Series
	G6	Harbor SEALs (Citizen Science)	Monitored	Point Time Series
DO	H1	NY State Department of Environmental Conservation, Division of Water Monitoring Data Portal	Monitored	Point Time Series
	H2	National Water Quality Monitoring Council	Monitored	Point Time Series
	H3	NYC DEP Harbor Survey	Monitored	Point Time Series
	H4	HRECOS (Hudson River Environmental Conditions Observing System)	Monitored	Point Time Series
	H5	Hudson River National Estuarine Research Reserve	Monitored	Point Time Series
	H6	Harbor SEALs (Citizen Science)	Monitored	Point Time Series

3.3 Built Environment

Table 3. Data Inventory - Built Environment Parameters.

	Ref ID	Data Sources	Data Type
NY Park Lands	J1	Parks Properties Map (NYC OpenData)	Polygon
	J2	Parks (NYCMAP / DCP)	Vector
	J3	New York State Parcel Data	Polygon
Maintained Navigational Channels	K1	National Channel Framework (eHydro)	Polygon
	K2	NOAA Electronic Navigational Charts	Mapped
CSO Outfall	L1	CSO Outfalls (NY DEC)	Points
	L2	Riverkeeper	Points

MS4 Outfall	M1	NYC MUNICIPAL SEPARATE STORM SEWER SYSTEM (MS4) MAP	Points and polygons
	M2	Municipal Storm Sewer System MS4	Polygons
	M3	Municipal Storm Sewer System MS4 Extended	Polygon
	M4	NYC DEP	Point Time Series
	M5	Westchester County MS4 Drainage Structure	Point
	M6	Westchester County GIS Municipal MS4 Non-Consortium Outfalls	Point

4. Data Review

Following development of the data inventory, the datasets were subjected to an evaluation to determine their applicability, reliability, and integration potential for subsequent analyses (**Phase I**). Each selected source was evaluated across six key criteria, and an overall score was calculated by averaging the individual scores across the evaluation criteria.

- Eval 1** Spatial coverage and applicability to target restoration areas.
- Eval 2** Temporal and spatial resolution.
- Eval 3** Data quality, accuracy, validation status, and known limitations.
- Eval 4** Data acquisition methods (e.g., instrumentation, modeling, in-situ monitoring).
- Eval 5** Processing methodology for model outputs and derived products if applicable and available.
- Eval 6** Technical requirements for integration into site selection workflows.

Due to the heterogeneous nature of the datasets evaluated, a standardized qualitative scoring system was applied across the six evaluation criteria to facilitate comparison among datasets. Data quality and usability were assessed using the rubric presented in **Table 4**.

Table 4. Scoring Rubric for Data Review.

Score	Meaning
-	Not Investigated
0	Not applicable/Not usable
1	Poor
2	Moderate
3	Good
4	Very good
5	Excellent

Table 5 summarizes the comparison matrix and associated evaluation scores for each data source selected for detailed assessment.

Table 5. Comparison Matrix Across Evaluation Criteria.

Data ID	Variable Category	Eval 1	Eval 2	Eval 3	Eval 4	Eval 5	Eval 6	Final Score
A4	Water/Current/Velocity/Streamflow	0	-	-	-	-	-	0.0
A5	Water/Current/Velocity/Streamflow	1	3	1	3	-	1	1.8
A7	Water/Current/Velocity/Streamflow	3	5	2	3	4	4	3.5
B3	Depth	2	3	4	4	4	2	3.2
B4	Depth	4	4	3	2	4	5	3.7
B7	Depth	5	3	2	3	4	5	3.7
C1	Wave exposure/fetch	5	5	4	4	4	5	4.5
D1	Shoreline Change Analysis	5	3	4	5	5	5	4.5
E1	Chl-a	0	-	-	-	-	-	0.0
E2	Chl-a	1	3	4	3	4	2	2.8
E3	Chl-a	2	3	3	3	4	2	2.8
E4	Chl-a	0	1	4	3	4	2	2.3
E7	Chl-a	4	4	1	4	4	3	3.3
F1	Salinity	0	-	-	-	-	-	0.0
F2	Salinity	3	3	4	4	4	4	3.7
F3	Salinity	4	3	3	4	4	4	3.7
F4	Salinity	0	1	4	4	4	2	2.5
G1	Turbidity or TSS	0	-	-	-	-	-	0.0
G2	Turbidity or TSS	1	3	4	3	4	2	2.8
G3	Turbidity or TSS	2	3	3	3	4	2	2.8
G5	Turbidity or TSS	0	1	4	3	4	2	2.3
H1	DO	0	-	-	-	-	-	0.0
H2	DO	1	3	4	4	4	2	3.0
H3	DO	2	3	3	4	4	2	3.0
H4	DO	0	1	4	4	4	2	2.5

Table 5, continued

J1	NYC Park Lands	3	5	3	5	5	5	4.3
J3	NYC Park Lands	5	5	5	5	5	5	5.0
K2	Maintained Navigational Channels	5	5	3	2	4	5	4.0
L1	CSO Outfall	5	5	5	5	5	5	5.0
M1	MS4 Outfall	-	-	-	-	-	-	-
M4	MS4 Outfall	4	2	4	5	5	5	4.2
M5	MS4 Outfall	2	1	4	5	5	5	3.7
M6	MS4 Outfall	2	1	1	1	1	5	1.8

Assessments of each data source were documented in Notion, a knowledge management platform. Detailed evaluation notes supporting the scoring are available here:

<https://natrx.notion.site/2d88fcbfc00f8096b35adf69beb72e34?v=2d88fcbfc00f80ce892f000c9e048da3>

5. Scoring Framework Development

This section describes the **Phase II Data Source Selection and Evaluation Methodology**.

5.1. Workshop Process and Governance

5.1.1. Workshop Overview

Phase II focused on translating the Phase I data inventory and evaluation results into a practical scoring methodology for site prioritization. Rather than developing the framework independently, Natrx facilitated a structured series of four collaborative workshops with BOP to jointly define scoring criteria, normalization approaches for each category variable, and final aggregation methodology (**Table 6**). These workshops were conducted to ensure methodological transparency, alignment with BOP priorities, and consistent interpretation and application of variables across sites. Prior to each workshop, Natrx and BOP exchanged peer-reviewed literature and technical guidance documents addressing oyster habitat requirements, water quality thresholds, and ecological performance metrics. This preparatory process ensured that all parameters, thresholds, and scoring approaches were scientifically grounded and aligned with established ecological criteria.

Table 6. Summary of Technical Workshops and Primary Focus Areas.

Workshop	Date	Participants	Primary Focus
#1	02/06/2026	Nick Brady (NatrX) Lise Montefiore (NatrX) Mike McCann (BOP) Carolyn Khoury (BOP)	Workshop organization, scoring framework structure, criteria categorization, and initial technical discussion on salinity
#2	02/13/2026	Nick Brady (NatrX) Lise Montefiore (NatrX) Mike McCann (BOP) Carolyn Khoury (BOP)	Continuation and completion of salinity and depth scoring; initiation of DO parameter development; evaluation of MS4 datasets and CSO-related variables.
#3	02/20/2026	Nick Brady (NatrX) Lise Montefiore (NatrX) Mike McCann (BOP) Carolyn Khoury (BOP) Cate Collinson (BOP)	Continuation and completion of DO parameter development; establishment of scoring approaches for Chl-a and turbidity; and integration of park-related variables into the overall site prioritization methodology.
#4	03/06/2026	Nick Brady (NatrX) Lise Montefiore (NatrX) Mike McCann (BOP) Carolyn Khoury (BOP) Cate Collinson (BOP)	Evaluation of water velocity datasets; recap of selected environmental variables and discussion of alternative score aggregation approaches for site prioritization.

5.1.2. Pre-Workshop Technical Preparation

Prior to the workshops, Natrx prepared supporting materials including a review of published Habitat Suitability Index (HSI) models for *C. virginica* (e.g., La Peyre et al. 2021; Beseres Pollack et al. 2012; Starke et al. 2011; Barnes et al. 2007) and evaluated how applicable these models were to the project's data landscape. These models provided useful reference points for suitability functions, ecological thresholds, and variable selections. However, following technical review, the team determined that a formal HSI framework was not appropriate for this engagement. HSI models are typically developed for specific regional contexts and depend on continuous, spatially comprehensive datasets, often derived from hydrodynamic modeling or dense monitoring networks, that exceed the scope and available data for this project.

Accordingly, the team adopted a prioritization-based scoring framework designed to generate relative site rankings using best-available datasets. This distinction is important: the objective of this effort is **not to produce absolute HSI, but to efficiently screen and rank candidate sites** to support planning and decision-making within realistic data constraints.

5.1.3. Variable Evaluation Framework and Normalization

Each category variable was reviewed through a structured workshop process. A standardized template was developed to ensure consistency and methodological transparency for each variable. The review framework addressed the following components:

- Biological relevance to *C. virginica*, including whether the variable functions primarily as a limiting factor, an enhancing factor, or both under different environmental conditions.
- Metric definition, including selection of appropriate statistical descriptors (e.g., mean, percentile, exceedance frequency), relevant seasonal windows, and temporal data ranges to reflect ecological response patterns.
- Data Processing and Spatial Assignment, including determination of vertical layer (surface vs. bottom), spatial assignment methodology, and identification of applicable Phase I datasets.
- Directionality of effects and scoring model selection, including use of threshold-based, ramp, optimum window, or exposure-based penalty functions, based on ecological response characteristics.
- Uncertainty and data confidence considerations, including assessment of data density, temporal representativeness, spatial coverage, and measurement limitations to prevent high uncertainty conditions from producing overly deterministic scoring outcomes.

Following variable evaluation and scoring function selection, variables were organized into two categories: **(1) scored variables** and **(2) flagged contextual variables**.

Scored variables include only water quality parameters for which numerical values were applied. These variables were **normalized to a common 0-1 scale to enable consistent comparison and aggregation across categories**. A score of 0 represents conditions least supportive of oyster survival and growth, while a score of 1 represents the most favorable conditions within the evaluated range. Scaling was performed using response functions (e.g., threshold, ramp, optimum window, or exposure-based penalty), ensuring that each variable's ecological behavior was preserved while translating raw measurements into a dimensionless score. These functions were defined collaboratively based on peer-reviewed literature and local expertise specific to New York Harbor conditions. This approach allows variables with different units, magnitudes, and ecological response shapes to be integrated into a composite framework without introducing scale bias.

Additionally, it was deemed useful to **report various statistics for some category variables along with the normalized score** for additional context. This recognizes the importance of subject-matter expertise while interpreting ranked scores.

Flagged contextual variables include physical parameters and aspects of the built environment that do not lend themselves to quantitative scoring but remain important for site suitability (e.g., proximity to park lands). These variables are not incorporated into the scoring framework and are instead **reported as contextual flags** in the ranking tables to support interpretation and decision-making.

5.2. Variable-Specific Scoring Development

The scoring framework is organized into three primary categories: Physical Parameters, Water Quality, and Built Environment. Variables within each category were reviewed following the workshop evaluation criteria. After discussion the final approach for each parameter was agreed upon by Natrx and BOP.

5.2.1. Physical Parameters

5.2.1.1. Depth

Depth is an important factor in restoration planning, as access, constructability, and long-term maintenance generally become more challenging in deeper water. At the site level, depth was summarized using the median value within each site polygon to reduce sensitivity to localized bathymetric variability or edge effects. The median depth is used as the primary metric in the scoring framework to capture the ecological relevance of depth conditions at the site scale. Depth was treated as a flagged binary metric: sites with a median depth ≤ 20 ft were identified for additional consideration, while deeper sites were not (**Table 7**).

Bathymetric data were derived from BlueTopo, built from NOAA's National Bathymetric Source. To additionally inform constructability and site planning, summary statistics describing the distribution of depths within each site were also evaluated. Specifically, each site was also characterized by the percentage of its area within three depth ranges (0-6 ft, 6-12 ft, and >12 ft). Sites may be described as shallow-dominant, mixed-depth, or deep-dominant based on the depth range representing the largest share of site area.

5.2.1.2. Velocity

Velocity was excluded as a direct parameter in the current analysis due to data limitations and the absence of clearly defined ecological thresholds. In addition, the practical value of including velocity as a separate parameter was considered limited in the context of the existing screening criteria. Many high-velocity environments (e.g., the East

River) were already excluded from consideration through other site constraints. Hydrodynamic influences related to water movement are expected to be captured indirectly through complementary analyses, including a previous larval transport modeling and wind fetch-based exposure metrics.

5.2.1.3. Wave exposure/fetch

Within the site prioritization framework, **wave exposure was treated as a contextual variable** that reflects both ecological function and implementation feasibility (**Table 7**). Moderate hydrodynamic energy supports water exchange and food delivery to oysters, while excessive exposure can reduce the stability of restoration structures. Wave exposure metrics were provided through *Natrx Assess*. Sites with higher wave height (≥ 3 ft) were flagged to indicate greater wave energy that may influence design requirements.

5.2.1.4. Shoreline Change Analysis (SCA)/Shoreline Erosion

Within the site prioritization framework, **shoreline change was identified as a contextual variable (Table 7)**. Shoreline change analysis was provided through *Natrx Assess*. Sites exhibiting erosion ≥ 1 ft were flagged to indicate dynamic shoreline conditions and to highlight locations where oyster restoration could provide shoreline stabilization benefits.

5.2.2. Water Quality Parameters

Within the Water Quality category, salinity, DO, and Chl-a were incorporated into the scoring framework based on their ecological relevance and the availability of datasets sufficient to support defensible scoring. Turbidity was also evaluated during the workshop but was ultimately excluded from the final scoring framework due to limited spatial and temporal coverage, data quality concerns, and insufficient discriminatory power across candidate sites. The available turbidity datasets did not consistently differentiate site conditions in a manner that would support meaningful relative ranking without introducing disproportionate uncertainty.

5.2.2.1. Salinity

Ecological Relevance

Salinity is a key ecological driver for *C. virginica*, affecting growth, feeding and filtration efficiency, reproduction timing, and disease susceptibility. Salinity functions as both a limiting factor at the low end of the gradient and an enhancing factor at moderate to higher concentrations, with ecological response dependent on magnitude, duration, and seasonal timing of exposure. Optimal growth generally occurs above 10 ppt, with severe

stress and mortality observed below 3-5 ppt (McFarland et al., 2018). Polyhaline environments have been shown to support substantially greater restoration effectiveness compared to mesohaline conditions, reflecting enhanced feeding and growth that outweigh increased predation and disease pressure (Smith & Castorani, 2023). Conversely, elevated summer salinity, often, driven by reduced freshwater inflow, can further intensify disease prevalence.

Although salinity is widely recognized as a critical driver of oyster physiology and restoration performance, defining universal threshold values is challenging because ecological responses are highly site-specific and influenced by local adaptation, hydrodynamics, and temporal variability. During the workshop, BOP presented salinity data collected at monitoring sites in the Hudson River (Tappan Zee region) where oyster populations are known to persist. The data indicated prolonged periods of low salinity, including 10th percentile values as low as 0.1-0.4 PSU and mean summer salinities between 3.4 and 4.8 PSU at sites supporting viable oyster populations. These observations highlighted the difficulty of applying the correct salinity thresholds and raised questions regarding the parameter's discriminatory power when comparing sites across similar estuarine environments. While the team acknowledged that salinity remained an important environmental variable it was recognized that its utility for differentiating candidate sites may be limited at the harbor-wide scale. Salinity may instead provide greater discriminatory resolution within localized spatial groupings (e.g., within Jamaica Bay). The team agreed to retain salinity in the scoring framework. However, key summary statistics (see *Metric Selection and Standardization Approach section*) will be reported alongside the derived score to provide contextual interpretation and avoid overreliance on a single composite metric.

Data Collection, Harmonization, and Processing

Salinity data were compiled from multiple monitoring programs, including the NYC DEP Harbor Survey (F3), the WQP dataset (F2), and HRECOS (F4). Records were harmonized to consistent units (PSU) and subjected to quality control procedures, including removal of implausible values, restriction to the ecologically relevant growing season (May-October), and filtering to retain recent observations from 2020 to present.

To ensure vertical consistency across datasets, salinity observations collected in waters deeper than 20 ft were excluded, as minimal stratification was observed under shallow conditions. The 20 ft threshold aligns with the depth range of candidate oyster restoration sites and typical reef placement elevations within the harbor, ensuring consistency between monitoring data and anticipated habitat conditions. This depth criterion was applied consistently across all data sources to maintain comparability.

For the WQP dataset, non-detect results were conservatively handled using half the reported detection limit when applicable. All salinity values across the different data sources were merged and stations with fewer than 10 seasonal observations were excluded to improve data robustness and reduce bias from sparsely sampled locations. However, due to data limitations, this threshold was relaxed to include stations with at least one observation in the lower Hudson water bodies upstream of Yonkers.

Because salinity observations were collected at discrete monitoring stations, estimating salinity conditions at candidate restoration sites and across unsampled areas required spatial interpolation. Spatial interpolation was performed for several salinity statistics (i.e., mean, median, 10th, and 90th percentiles) using an inverse distance weighting (IDW) approach based on hydrologically constrained (in-water) distances. Distances between locations were calculated along connected water pathways using a raster-based representation of the water domain. Movement was restricted to water cells, and least-cost path distances were computed using Dijkstra's algorithm to ensure that interpolation reflects the true geometry of the estuarine system and prevent influence from stations separated by land barriers.

For each prediction location, salinity was estimated using the k nearest hydrologically connected stations, with weights defined as the inverse of distance raised to a power parameter. The number of neighboring stations (k) and the distance-decay exponent were optimized through sensitivity analysis using a grid search framework. Model performance was evaluated using leave-one-out cross-validation (LOO-CV), and the optimal parameter combination was selected based on minimization of root mean square error (RMSE).

Metric Selection and Standardization Approach

The primary salinity scoring metric is the mean salinity calculated over the May-October seasonal window using data from the most recent six-year period available (2020-2025) (**Table 7**). In addition to the derived score, supporting distributional statistics, including the 10th and 90th percentiles, and seasonal median, were reported for each site to characterize lower-bound stress exposure and upper-bound salinity conditions and to support interpretive transparency.

Mean salinity values were transformed using the suitability function published by Starke et al. (2011), developed for *C. virginica* restoration within the Hudson River watershed. The function follows an optimum window response shape: suitability is zero at approximately 5 PSU, increases steeply to roughly 0.89 near 12 PSU, remains high (approximately 0.85-0.89) through 20 PSU, and declines toward zero at 40 PSU (**Figure 1**). Given its local development and ecological relevance, this function provides a defensible basis for scoring. Resulting suitability values were normalized to a 0-1 scale to ensure consistency within the broader prioritization framework.

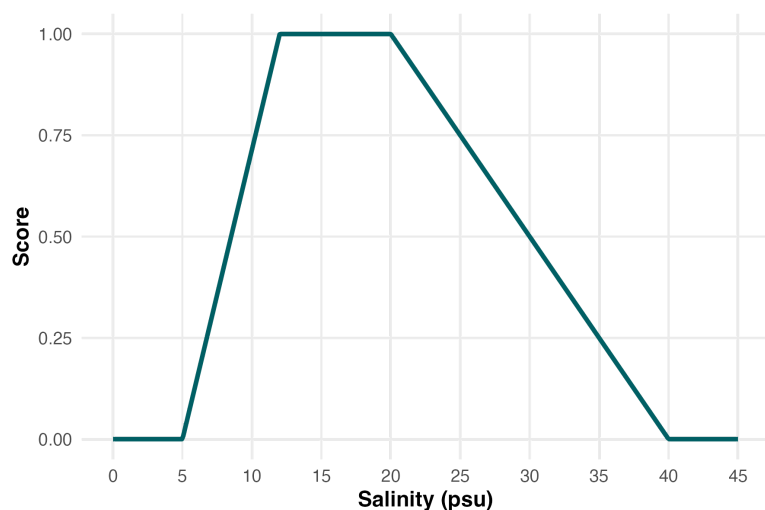


Figure 1. Salinity Suitability Transformation from Mean Salinity.

Uncertainty Quantification

IDW is an interpolation method that does not provide an intrinsic estimate of prediction uncertainty. Uncertainty was therefore quantified using a hierarchical bootstrap approach. Bootstrap distributions were used to derive site-level uncertainty statistics, including standard errors and 95% confidence intervals (CIs). Confidence classifications based on CI width were assigned as follows

- High when $CI \leq 6$ PSU;

- Moderate when $6 < CI \leq 10$;
- Low when $CI > 10$.

5.2.2.2. Chl-a

Ecological Relevance

Chl-a is a proxy for phytoplankton biomass and overall system productivity. For *C. virginica*, phytoplankton availability directly influences food supply, growth rates, and energy allocation. However, the ecological relationship is non-linear. Very low Chl-a concentrations ($< 5 \mu\text{g}\cdot\text{L}^{-1}$) are generally associated with low-productivity conditions that may limit food availability and constrain growth. Moderate concentrations (approximately $5\text{--}20 \mu\text{g}\cdot\text{L}^{-1}$) are typically characteristic of mesotrophic systems that support adequate feeding and growth. Elevated concentrations ($> 20 \mu\text{g}\cdot\text{L}^{-1}$) can enhance food availability for oysters but are also commonly associated with eutrophic conditions where algal blooms, increased turbidity, and subsequent oxygen depletion are more likely. However, the ecological response to Chl-a is not linear. In urbanized estuarine systems such as New York Harbor, seasonal Chl-a concentrations can exceed $30 \mu\text{g}\cdot\text{L}^{-1}$ while still supporting oyster persistence. Ecological outcomes depend on bloom duration, phytoplankton community composition, hydrodynamic conditions, rather than a single concentration threshold.

Data Collection, Harmonization, and Processing

Chl-a data were compiled from the NYC DEP Harbor Survey (E3) and the WQP (E2). During review, paired samples showed inconsistencies between laboratory-analyzed and sensor-derived Chl-a measurements, including differences in magnitude and variability. To maintain methodological consistency and avoid cross-method bias, only laboratory-analyzed Chl-a data were retained for this analysis. As a result, sensor-based datasets, including HRECOS records, were excluded.

The analysis was limited to the most recent six-year period available (2020-2025) and restricted to the May-November seasonal window to align with periods when water temperatures are generally above approximately 10°C and oyster filtration is not thermally inhibited.

All retained observations were harmonized to consistent units ($\mu\text{g}\cdot\text{L}^{-1}$) and screened for quality, including removal of implausible values and records lacking valid spatial coordinates. To ensure vertical consistency and relevance to near-surface phytoplankton dynamics, analysis was restricted to samples collected within the upper 6.6 ft (2 m) of the water column; most were surface or grab samples.

Unlike salinity, Chl-a was not interpolated, as high variability and limited spatial structure did not support robust modeling. Basin-scale interpolation would introduce artificial smoothing across heterogeneous waterbodies and attenuate localized productivity gradients and bloom dynamics relevant to oyster feeding.

Accordingly, station-level summary statistics (i.e., mean, median, and 10th and 90th percentiles) were computed and used within a distance-weighted pooling framework. For each candidate site, Chl-a conditions were estimated by aggregating information from nearby monitoring stations using hydrologically constrained (in-water) distances. Distances between locations were calculated along connected water pathways using a raster-based representation of the water domain, where movement was restricted to water cells and shortest-path distances were computed using Dijkstra's algorithm. This ensures that station influence reflects the true geometry of the estuarine system and prevents influence from stations separated by land.

Station contributions were weighted as a function of distance using an inverse distance decay formulation, with weights normalized across all contributing stations, such that closer stations exert greater influence on the estimates. A primary search radius (5 km) was used to identify nearby stations, with an extended radius (10 km) applied where necessary to ensure sufficient data coverage. The distance-decay exponent was optimized through LOO-CV to minimize estimation error. Distance-weighted aggregation of station-specific mean, median, and percentile metrics was used to characterize Chl-a conditions at each site.

Metric Selection and Standardization Approach

The primary scoring metric for Chl-a was the seasonal median concentration calculated over the May-November window (**Table 7**). The median was selected instead of the mean to reduce sensitivity to short-duration bloom events and extreme values that may not reflect persistent trophic conditions. In addition to the derived score, the 10th and 90th percentiles were reported for each site to characterize overall productivity and potential bloom intensity.

Chl-a score was derived using a linear normalization approach to represent relative trophic support for oyster feeding. A Chl-a concentration of 0 $\mu\text{g}\cdot\text{L}^{-1}$ corresponds to a score of 0, indicating no available phytoplankton biomass. The score increased linearly with Chl-a concentrations, with the maximum observed median Chl-a value across sites assigned a score of 1.0 (**Figure 2**). Intermediate values were scaled proportionally between these bounds. This approach was used to discriminate sites based on relative differences in food availability.

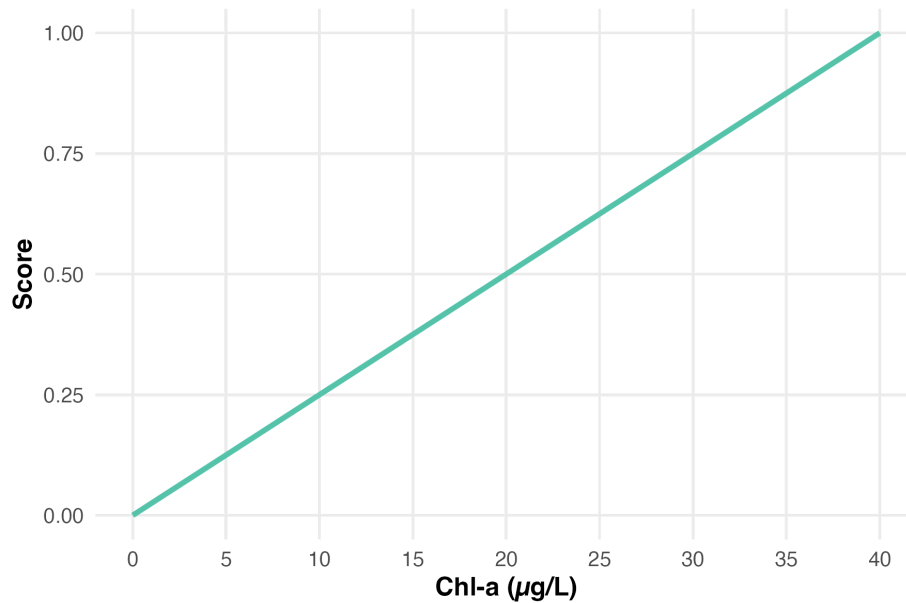


Figure 2. Chl-a Scoring Transformation.

Uncertainty Quantification

Uncertainty in Chl-a statistics was evaluated along two complementary dimensions: (1) statistical uncertainty and (2) spatial data support.

Statistical uncertainty was quantified using a hierarchical bootstrap framework ($n = 1,000$) that resampled both monitoring stations and temporal observations (years) within stations. For each bootstrap iteration, station-level metrics were recomputed and propagated through the same distance-weighted aggregation framework, generating a distribution of site-level estimates for each candidate location. Summary statistics and 95% CIs were then derived from the bootstrap distributions.

Chl-a uncertainty was classified based on the CI width of the weighted median Chl-a estimate as follows:

- High confidence when $CI \leq 10 \mu\text{g.L}^{-1}$;
- Moderate confidence when $10 < CI \leq 20 \mu\text{g.L}^{-1}$;
- Low confidence when $CI > 20 \mu\text{g.L}^{-1}$.

Spatial data support was characterized using the hydrologically constrained distance to the nearest monitoring station. Sites located closer to monitoring stations were considered better supported by observed data, while sites farther from stations should be interpreted with greater caution. Spatial confidence ranges were classified as follows:

- High confidence when distance $\leq 300 \text{ m}$;
- Moderate confidence when $300 < \text{distance} \leq 1,000 \text{ m}$;

- Low confidence when distance > 1,000 m.

5.2.2.3. DO

Ecological Relevance

DO is a critical environmental parameter governing oyster survival, growth, and metabolic function. Oxygen availability directly influences respiration, filtration activity, energy allocation, and reproductive processes. Concentrations below $1 \text{ mg}\cdot\text{L}^{-1}$ are considered lethal hypoxia and have been associated with direct mortality. Levels below approximately $3 \text{ mg}\cdot\text{L}^{-1}$ induce severe physiological stress, prompting valve closure and inhibition of feeding and filtration. Values between approximately 3 and $5 \text{ mg}\cdot\text{L}^{-1}$ are generally considered suboptimal, with reductions in growth and reproductive output. Concentrations above approximately $5 \text{ mg}\cdot\text{L}^{-1}$ are typically suitable for normal oyster function. Thus, DO functions primarily as a limiting factor: ecological response is characterized by sharp declines in suitability below defined thresholds instead of progressive improvement at higher concentrations.

Data Collection, Harmonization, and Processing

DO data were compiled from multiple monitoring programs, including the NYC DEP Harbor Survey (H3), the WQP dataset (H2), and HRECOS (H4). Records were harmonized to consistent units ($\text{mg}\cdot\text{L}^{-1}$) and screened for quality, including removal of values outside the plausible range ($0\text{-}20 \text{ mg}\cdot\text{L}^{-1}$) and exclusion of records without valid spatial coordinates. To focus on periods of highest hypoxia risk and greatest ecological relevance for oyster stress, the analysis was restricted to the June-September seasonal window over the most recent six-year period (2020-2025).

For the NYC DEP dataset, DO values were standardized using a hierarchical approach that prioritizes bottom measurements, followed by surface values when bottom data were unavailable. Differences between surface and bottom measurements were small (median difference: -0.08 to $-0.26 \text{ mg}\cdot\text{L}^{-1}$), indicating limited vertical stratification at the sampled sites. When only Winkler titration measurements were available, bottom values were prioritized over surface values, and a correction of $-0.49 \text{ mg}\cdot\text{L}^{-1}$ was applied to account for the observed bias relative to paired CTD samples. For the WQP dataset, records were filtered to the standardized DO parameter, restricted to units of $\text{mg}\cdot\text{L}^{-1}$, and limited to samples collected at depths ≤ 20 ft to maintain consistency with shallow habitat conditions.

Station-level summaries were then computed, including mean DO, median DO, and the frequency of low-oxygen exposure expressed as the percentage of observations below $3 \text{ mg}\cdot\text{L}^{-1}$.

DO observations were collected at discrete monitoring stations and exhibited substantial spatial variability. Due to data sparsity and the localized nature of hypoxic conditions, continuous spatial interpolation approaches were not considered appropriate. Instead, station-level metrics were used within a distance-weighted pooling framework.

For each candidate site, DO conditions were estimated by aggregating information from nearby monitoring stations using hydrologically constrained (in-water) distances. Distances between locations were calculated along connected water pathways using a raster-based representation of the water domain, where movement was restricted to water cells and shortest-path distances were computed using Dijkstra's algorithm. This approach ensures that station influence reflects the true geometry of the system and prevents influence from stations separated by land.

Station contributions were weighted as a function of distance using an inverse distance decay formulation, with weights normalized across all contributing stations. A primary search radius (5 km) was used, with an extended radius (10 km) applied where necessary to ensure sufficient data coverage. The distance-decay exponent was optimized through LOO-CV to minimize estimation error. Distance-weighted aggregation of station-specific mean DO, median DO, and hypoxic exposure metrics (percentage of observations below 3 mg·L⁻¹) was used to characterize site-level oxygen conditions.

Metric Selection and Standardization Approach

DO score was evaluated using an exposure-based scoring approach that incorporates uncertainty in hypoxic conditions (**Table 7**). The metric was based on the upper bound of the 95% CI of the percentage of observations below 3 mg·L⁻¹, representing a conservative estimate of low-oxygen exposure over the June-September seasonal window using data from the most recent six-year period available. Sites with very low hypoxia risk ($\leq 2\%$) were assigned a high suitability score (1.0), while sites with moderate exposure (2-10%) received a reduced score (0.75). Sites with higher potential hypoxia exposure ($> 10\%$) were assigned a lower suitability score (0.5), reflecting increased ecological risk. A reduced score of 0.5 (rather than 0) was applied to avoid fully penalizing sites based on potentially rare or short-duration hypoxic events (**Figure 3**). This approach acknowledges that episodic low-oxygen conditions may occur without precluding overall site suitability, while still recognizing meaningful ecological risk within the prioritization framework.

In addition to the derived score, the percentage exceedance value and supporting distributional statistics (e.g., median and percentile-based metrics) were reported for each site to characterize variability in oxygen conditions and support interpretive transparency.

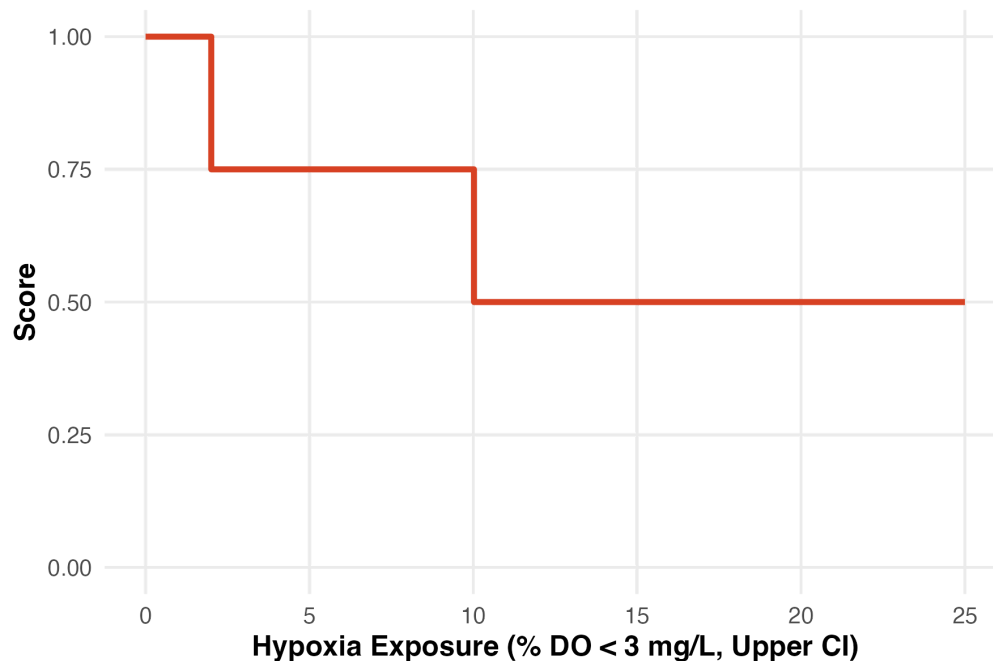


Figure 3. Hypoxia Exposure Scoring Function.

Uncertainty Quantification

Uncertainty in DO statistics was evaluated along two complementary dimensions: (1) statistical uncertainty and (2) spatial data support.

Statistical uncertainty was estimated to account for sampling variability across both monitoring stations and years. Uncertainty was quantified using a hierarchical bootstrap approach ($n = 1,000$) that resamples both monitoring stations and temporal observations (years) within stations. Uncertainty was incorporated directly into the scoring framework by using the upper bound of the bootstrap-derived 95% confidence interval for hypoxic exposure

Spatial data support was characterized using the hydrologically constrained distance to the nearest monitoring station. Sites located closer to monitoring stations were considered better supported by observed data, while sites farther from stations should be interpreted with greater caution. Spatial confidence ranges were classified as follows:

- High confidence when distance ≤ 300 m;
- Moderate confidence when $300 < \text{distance} \leq 1,000$ m;
- Low confidence when distance $> 1,000$ m.

5.2.3. Built Environment Parameters

5.2.3.1. NYC Parklands

Parkland proximity was identified as a contextual variable (Table 7); relevant to permitting and implementation instead of an ecological performance scoring. The NYC Parks dataset was selected as the primary spatial reference, with state parcel codes (e.g., J3) reviewed but used as supplementary information. A proximity-based approach was adopted to determine whether a site is considered adjacent to park lands. **Candidate sites with park lands within 1,640 ft (500 m) were designated as park land sites.** Calculations were performed using a spatial join and buffer around the sites. Sites meeting this criterion were flagged; others were not assigned this designation.

5.2.3.2. MS4 Outfalls

Proximity to **MS4 outfalls was treated as a contextual variable** within the site prioritization framework (**Table 7**). To maximize spatial completeness, MS4 data were compiled from NYCDEP and supplemented with Westchester Consortium and non-consortium datasets, then merged into a unified layer. The flagged metric was based on a threshold rule: **sites within 50 ft were flagged** for additional consideration.

5.2.3.3. CSO Outfalls

Proximity to CSO outfalls was treated as a contextual variable within the site prioritization framework (**Table 7**). Available NYC CSO mapping was used to develop a spatial layer of outfall locations across the project area. The flagged metric was based on a threshold rule: **sites within 50 ft were flagged** for additional consideration.

5.3. Scoring Parameter Summary

Table 7 summarizes the scoring framework developed through the workshops. For each parameter, the table indicates whether the variable was scored or flagged, the scoring rule or treatment applied, the primary evaluation metric, any supporting statistics reported for context, and whether a spatial or temporal uncertainty layer accompanies the metric.

Table 7. Scoring Parameters Summary. **Scored**=contributes a 0-1 suitability value to the composite index. **Flagged** = binary contextual indicator; does not penalize or alter composite score. **Uncertainty Layer** = spatial confidence tier and distance to nearest proxy station reported alongside score.

Parameter	Treatment	Scoring Method/ Rule	Score Range	Primary Metric	Supporting statistics	Uncertainty Layer
Physical Parameters						
Depth	Flagged	Binary cutoff at 20 ft, 1 at ≤ 20 ft; 0 at > 20 ft.	Flag (Y/N)	Median site depth	% area in depth ranges (0–6 ft, 6–12 ft, >12 ft); depth class (shallow-dominant, mixed, deep-dominant)	No
Wave Exposure	Flagged	Contextual flag for high-fetch sites (wave height ≥ 3 ft)	Flag (Y/N)	Fetch / wave energy	-	No
Shoreline Change	Flagged	Visual flag where erosion exceeds defined 1 ft/year; not penalized in score	Flag (Y/N)	Erosion Rate	-	No
Water Quality Parameters						
Salinity	Scored	Starke et al. (2011) optimum-window function; 0 at 5 PSU, peak 12–20 PSU, 0 at 40 PSU	0-1	Seasonal mean (May-Oct, 6-yr)	P10, P90, mean	Yes

Chl-a	Scored	Linear scaling (0-1) based on 0 to max observed Chl-a	0-1	Seasonal median (May-Nov, 6-yr)	P10, P90, median	Yes
DO	Scored	Threshold-based scoring using hypoxia exposure ($\text{DO} < 3 \text{ mg}\cdot\text{L}^{-1}$, upper 95% CI): 1.0 ($\leq 2\%$), 0.75 (2–10%), 0.5 ($> 10\%$)	0.5/0.75/1.0	% seasonal measurements $< 3 \text{ mg}\cdot\text{L}^{-1}$ (June–September, 6-yr)	% $< 3 \text{ mg}\cdot\text{L}^{-1}$, P10, P90, median	Yes
Built Environment Parameters						
Parkland Proximity	Flagged	Distance to nearest parkland boundary; flagged if within 1,640 ft (500 m) buffer	Flag (Y/N)	Distance to nearest parkland boundary (m)	-	No
MS4 Outfalls	Flagged	Threshold rule: sites within 50 ft of an MS4 outfall are flagged for additional consideration	Flag (Y/N)	Distance to nearest MS4 outfall (ft)	-	No
CSO Outfalls	Flagged	Threshold rule: sites within 50 ft of a CSO outfall are flagged for additional consideration	Flag (Y/N)	Distance to nearest CSO outfall (ft)	-	No

5.4. Composite Scoring and Aggregation Framework

5.4.1. Water Quality Scoring Aggregation

To aggregate individual water quality parameter scores into an overall site suitability metric for oyster restoration, a DO-modifier formulation was selected as the primary approach, with additional methods used to evaluate ranking robustness.

The DO-modifier approach was defined as:

$$((\text{salinity score} + \text{Chl-a score}) / 2) \times \text{DO score}$$

Under this formulation, salinity and Chl-a represent baseline environmental conditions supporting oyster habitat, while DO acts as a limiting modifier that reduces site scores under hypoxic conditions.

To evaluate the sensitivity of site rankings to the aggregation method, two alternative formulations were also applied: (1) an arithmetic mean and (2) a geometric mean. The arithmetic mean computes the composite score as the average of salinity, Chl-a, and DO scores, allowing full compensatory behavior among parameters. In contrast, the geometric mean combines scores through multiplication, resulting in a more restrictive formulation in which low values in any parameter exert a stronger suppressive effect on the final score.

No explicit weighting was applied to individual parameters. Assigning weights would require a strong empirical basis to justify differential importance among variables. A transparent, unweighted framework was therefore adopted to support interpretability and facilitate communication with stakeholders.

Rankings produced by these alternative approaches were compared to those obtained using the DO-modifier formulation to assess robustness.

5.4.2. Overall Confidence Classification

Overall confidence was derived by combining multiple individual confidence dimensions into a single integrated assessment. Each site was evaluated across four contributing confidence components (**Table 8**; See **Section 6.2.2. Uncertainty Quantification**), and the final classification reflects the combined assessment of these components instead of any single metric.

Table 8. Summary of Confidence Assessment Methods Developed for Water Quality Parameters.

Parameter	Confidence (from Statistical Uncertainty)	Spatial Data Support (Distance to Nearest Station)
Salinity	High: CI width $\leq$ 6 PSU Moderate: 6-10 PSU Low: > 10 PSU	NA
Chl-a	High: CI width $\leq$ 10 $\mu\text{g.L}^{-1}$ Moderate: 10-20 $\mu\text{g.L}^{-1}$ Low: > 20 $\mu\text{g.L}^{-1}$	High: $\leq$ 300 m Moderate: 300–1,000 m Low: > 1,000 m
DO	Indirectly reflected in the metric (upper 95% CI bound of hypoxic exposure)	High: $\leq$ 300 m Moderate: 300–1,000 m Low: > 1,000 m

The overall confidence classification follows a rule-based aggregation framework (**Table 9**):

Table 9. Rule-Based Aggregation of Component Confidence into Overall Confidence.

Overall Confidence	Combination of component-level confidence
High	4 High; or 3 High + 1 Moderate
Moderate+	3 High + 1 Low; 2 High + 2 Moderate; 2 High + 1 Moderate + 1 Low
Moderate	4 Moderate; 2 High + 2 Low; 1 High + 2 Moderate + 1 Low
Moderate-	3 Moderate + 1 Low; 2 Moderate + 2 Low; 1 High + 1 Moderate + 2 Low
Low	4 Low; 3 Low + 1 Moderate; 3 Low + 1 High

5.4.3. Other Parameters

All remaining parameters (e.g., depth, proximity to park) were summarized based on the assigned flags (**Table 7**).

6. Site Analysis and Scoring Implementation

This section describes the **Phase III Site Analysis and Scoring Implementation**.

6.1. Natrx Assess Data Enrichment

6.1.1. Erosion Data

See Appendix A for Methodology- Natrx Assess - Shoreline Change Analysis (SCA). Some sites were not processed using the Natrx Assess platform because they fall outside the scope of the analysis (e.g., fully hardened shorelines).

6.1.2. Wind-Wave Data

See Appendix B for Methodology: Natrx Assess - Wind-driven Wave Modeling. Some sites were not processed using the Natrx Assess platform because they fall outside the scope of the analysis (e.g., fully hardened shorelines).

6.2. Data Integration and Site Scoring

6.2.1. Physical Parameters

Reported distributions for the following physical parameters are summarized below:

- **Depth:** Of the 78 sites, 68 have more than 50% of their area within the 0-20 ft depth range (score = 1).
- **Erosion:** Of the 78 sites, 74 were analyzed using Natrx Assess. Among these, 23 exhibit erosion, including 8 sites with erosion rates exceeding 1 ft/year.
- **Wave Exposure:** Of the 78 sites, 74 were analyzed using Natrx Assess. Among these, 22 experience wave heights exceeding 3 ft at some frequency.

6.2.2. Water Quality

6.2.2.1. Summary of Water Quality Metrics

Salinity

Salinity ranged from 7 PSU (e.g., Hudson River) to 30 PSU (e.g., Jamaica Bay), with a median of 25 PSU. CIs were narrow for the majority of sites (CI width < 6 PSU), indicating high certainty. In contrast, sites in the Hudson River exhibited moderate to low certainty, with wider CI (7 to 17 PSU), likely due to a lower density of monitoring stations and observations.

Chl-a

Chl-a medians ranged from 1 $\mu\text{g}\cdot\text{L}^{-1}$ (Upper Harbor) to 30 $\mu\text{g}\cdot\text{L}^{-1}$ (Jamaica Bay), with an overall median of 14 $\mu\text{g}\cdot\text{L}^{-1}$.

DO

The percentage of observations where DO fell below 3 $\text{mg}\cdot\text{L}^{-1}$ ranged from 1% to 20%. However, this metric exhibited substantial uncertainty, with CI widths reaching up to 78% and a median CI width of 12%. The upper bound of the CI was used to represent the percentage of observations below 3 $\text{mg}\cdot\text{L}^{-1}$, providing a more conservative estimate that accounts for potential underestimation of low-oxygen conditions.

6.2.2.2. Evaluation of Scoring Aggregation Methods

The DO-modifier aggregation approach was applied as the primary method and compared against two alternative aggregation methods: (1) an arithmetic mean and (2) a geometric mean. Results showed consistent rankings for the highest- and lowest-priority sites across all methods. However, some mid-ranked sites exhibited greater sensitivity to the choice of aggregation framework (**Figure 1C**, Appendix C).

These results support the use of the DO-modifier approach as the primary ranking method, while providing confidence that key prioritization outcomes are robust and not driven by the selection of aggregation methods.

6.2.2.3. Site Ranking

The scoring framework identified several sites with the highest rankings in terms of water quality. The top-ranked sites include Arthur Kill and multiple locations within the Living Breakwaters system, followed by Wolfe's Pond, Conch Basin, and Arthur Kill-Pralls River.

In contrast, the lowest-ranked sites were City Island, Powell's Cove, Malba, and multiple locations within the Bush Terminal area, as well as Earle and Ossining.

However, while many sites are ranked highly, the data supporting some of these locations are comparatively limited, resulting in lower confidence relative to other sites with more robust observational coverage. As illustrated in **Figure 4**, the distribution of confidence levels varies across DO-modifier rank groups. The highest-ranked sites (1-20) show a substantial proportion of low-confidence classifications, indicating that many top-ranked locations are supported by relatively limited or uncertain data. In contrast, mid-ranked groups (21-40 and 41-60) exhibit a greater share of moderate to high confidence, suggesting more robust data support in these ranges. The lowest-ranked sites (61-78) are again characterized by a higher proportion of low-confidence classifications, although some moderate-confidence sites are also present.

Overall, this pattern indicates that data confidence is not uniformly distributed across rankings, with both the highest- and lowest-ranked groups showing greater uncertainty compared to mid-ranked sites.

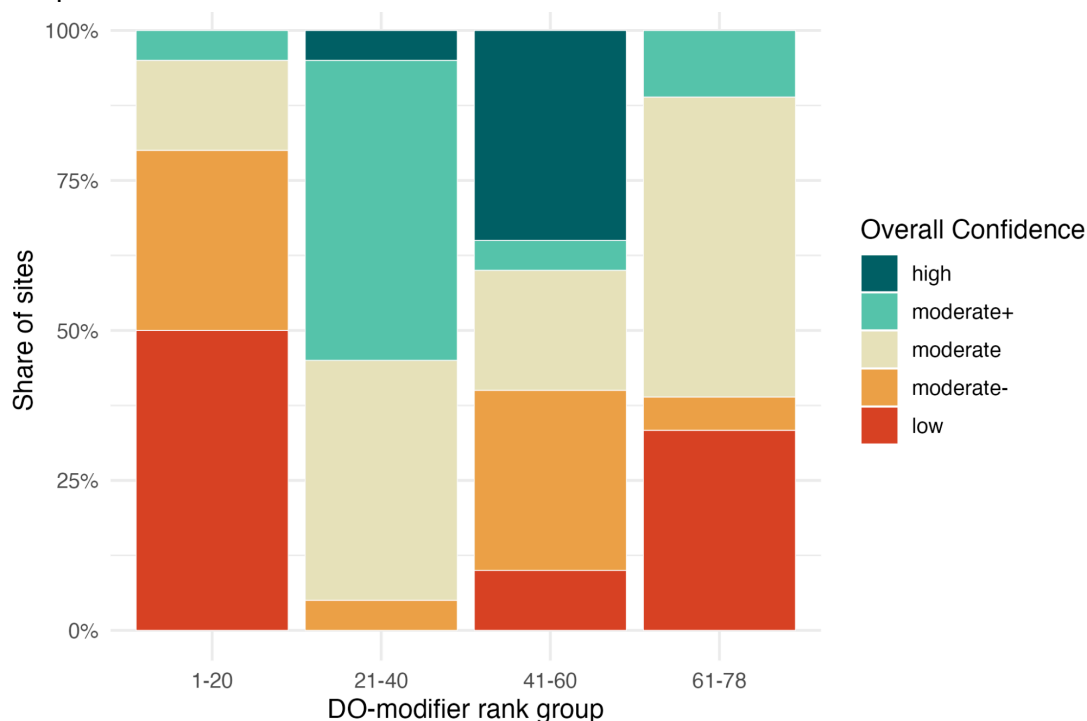


Figure 4. Confidence Composition by DO-modifier Rank Group.

6.2.3. Built Environment

MS4

Seven sites are located within 50 ft of an MS4 (**Table 10**).

Table 10. Sites Located Within 50 ft of a MS4.

ID	Site	Nearest MS4 (ft)
69	Paerdegat Basin	0
68	Paerdegat Basin	0
23	Bayswater Park - NYC Parks	0
27	Arthur Kill	0
4	Jamaica Bay Park 1	18
53	Brant Point Wildlife Sanctuary	37
13	Fort Totten	49

The remaining sites are located at distances ranging from 53 ft to over 40,000 ft.

CSO

Four sites are located within 50 ft of a CSO (**Table 11**).

Table 11. Sites Located Within 50 ft of a CSO.

ID	Site	Nearest CSO (ft)
2	Williamsburg River Ring	0
35	Bush Terminal - USACE	0
34	Bush Terminal - USACE	0.40
55	Bushwick Inlet	45

The remaining sites are located at distances ranging from 160 to over 57,000 ft.

NYC Parklands

Of the 78 sites, 59 are within 1,640 ft (500 m) of NYC parklands.

7. Site Prioritization

7.1. Final Site Rankings and Confidence Interpretation

Final site rankings are presented across four tables (**Tables 12-15**), organized from best to worst water quality conditions for oyster suitability (ranks 1-20, 21-40, 41-60, and 61-78), with built environment and physical parameters reported as flagged contextual variables.

Final site rankings were derived from the composite DO-modifier score, rounded to two decimal places to avoid spurious differentiation among sites whose underlying scores differ only at a trivial precision. Sites with identical rounded scores were further ordered by the overall confidence classification (high > moderate+ > moderate > moderate- > low), ensuring that, between otherwise equivalent sites, the one supported by stronger data received the better rank.

High-ranking sites were not always associated with high confidence, indicating that ecological suitability and data support do not fully align. Rankings should therefore be interpreted together with the confidence layer: sites with high rank and high confidence are the strongest candidates, sites with high rank but low confidence are promising but uncertain, and sites with low rank and low confidence are both weakly ranked and poorly supported.

Table 12. Summary of Site Rankings and Associated Environmental and Physical Conditions (Ranks 1-20).

ID	Site	Water Quality Parameters		Physical Parameters			Built Environment		
		Rank	Score	Wave Exposure	Erosion	Depth	Near CSO	Near MS4	Near Park
27	Arthur Kill	1	0.87	No	Yes	No	Yes	Yes	Yes
36	Living Breakwaters	2	0.79	No	No	Yes	Yes	NA	Yes
37	Living Breakwaters	2	0.79	No	No	Yes	Yes	NA	Yes
38	Living Breakwaters	4	0.78	No	No	Yes	Yes	NA	Yes
40	Living Breakwaters	5	0.76	No	No	Yes	Yes	NA	Yes
41	Living Breakwaters	5	0.76	No	No	Yes	Yes	NA	Yes
42	Living Breakwaters	7	0.74	No	No	Yes	Yes	NA	Yes
30	Wolfe's Pond	8	0.65	No	No	Yes	Yes	No	Yes
24	Conch Basin	9	0.62	No	No	Yes	Yes	No	No
28	Arthur Kill - Pralls River	10	0.62	No	No	Yes	Yes	Yes	Yes
25	Arthur Kill	10	0.62	No	No	Yes	Yes	Yes	Yes
26	Arthur Kill	10	0.62	No	No	Yes	Yes	No	Yes
52	Dubos Point Wildlife Sanctuary	13	0.59	No	No	Yes	Yes	No	No
29	Arthur Kill - Pralls River	13	0.59	No	No	Yes	Yes	Yes	Yes
21	Barbadoes Basin	15	0.57	No	No	Yes	Yes	No	Yes
58	Yonkers - South	16	0.56	No	No	No	NA	NA	Yes
59	Yonkers - North	17	0.56	No	No	No	NA	NA	No
60	Hastings-on-Hudson	17	0.56	No	No	No	NA	NA	Yes
7	Little Neck Bay	19	0.54	No	No	Yes	NA	NA	Yes
31	Save the Sound Big Rock	20	0.52	No	No	Yes	Yes	No	Yes

***Confidence Layer of Water Quality Score:**

High
 Moderate +
 Moderate
 Moderate -
 Low

Table 13. Summary of Site Rankings and Associated Environmental and Physical Conditions (Ranks 21-40).

ID	Site	Water Quality Parameters		Physical Parameters			Built Environment		
		Rank	Score	Wave Exposure	Erosion	Depth	Near CSO	Near MS4	Near Park
8	Joe Michael Mile	21	0.5	No	No	Yes	NA	NA	Yes
16	Brooklyn Marine Park	22	0.5	No	No	Yes	Yes	No	Yes
17	Brooklyn Marine Park	23	0.49	No	No	Yes	Yes	No	Yes
6	Alley Creek	24	0.49	No	No	Yes	Yes	Yes	Yes
10	Orchard Beach	25	0.47	No	No	Yes	NA	Yes	Yes
11	Orchard Beach	25	0.47	No	No	Yes	Yes	Yes	Yes
12	Orchard Beach	25	0.47	No	No	Yes	Yes	Yes	Yes
23	Bayswater Park - NYC Parks	25	0.47	No	Yes	Yes	Yes	Yes	Yes
9	Hart Island	29	0.47	No	No	Yes	Yes	Yes	Yes
22	Bayswater Park - HPD	30	0.46	No	No	Yes	Yes	Yes	Yes
13	Fort Totten	30	0.46	No	Yes	Yes	NA	NA	Yes
5	Jamaica Bay Park 2	30	0.46	No	No	Yes	Yes	Yes	Yes
4	Jamaica Bay Park 1	33	0.45	No	Yes	Yes	Yes	No	Yes
46	Shirley Chisholm State Park	33	0.45	No	No	No	Yes	No	Yes
54	Vernam Barbadoes	35	0.44	No	No	Yes	Yes	No	Yes
53	Brant Point Wildlife Sanctuary	36	0.44	No	Yes	Yes	Yes	Yes	Yes
51	Rockaway Community Park	37	0.44	No	No	Yes	Yes	Yes	Yes
76	Brooklyn Bride Park	38	0.43	No	No	Yes	NA	NA	No
18	Brooklyn Marine Park	39	0.42	No	No	Yes	Yes	Yes	Yes
69	Paerdegat Basin	40	0.41	No	Yes	Yes	NA	Yes	Yes

* Confidence Layer of Water Quality Score:

High Moderate + Moderate Moderate - Low

Table 14. Summary of Site Rankings and Associated Environmental and Physical Conditions (Ranks 41-60).

ID	Site	Water Quality Parameters		Physical Parameters			Built Environment		
		Rank	Score	Wave Exposure	Erosion	Depth	Near CSO	Near MS4	Near Park
68	Paerdegat Basin	41	0.4	No	Yes	Yes	NA	Yes	Yes
67	Paerdegat Basin	42	0.37	No	No	Yes	NA	Yes	Yes
55	Bushwick Inlet	43	0.37	Yes	No	Yes	NA	NA	Yes
2	Williamsburg River Ring	43	0.37	Yes	No	Yes	NA	NA	No
56	Bushwick Inlet	45	0.37	No	No	Yes	NA	NA	Yes
66	Paerdegat Basin	46	0.36	No	No	Yes	NA	Yes	Yes
50	Four Sparrow	47	0.33	No	No	Yes	Yes	Yes	Yes
19	Lemon Creek	47	0.33	No	No	Yes	NA	Yes	Yes
61	Dobbs Ferry	49	0.33	No	No	No	NA	NA	Yes
84	Flushing Bay	50	0.32	No	No	Yes	NA	NA	Yes
82	Flushing Bay	51	0.31	No	No	Yes	NA	NA	Yes
83	Flushing Bay	51	0.31	No	No	Yes	NA	NA	Yes
3	Bergen Basin	53	0.31	No	No	No	Yes	No	Yes
44	Head of Bay - USACE	54	0.3	No	No	Yes	Yes	NA	No
49	Fresh Creek Basin	55	0.29	No	No	Yes	NA	Yes	Yes
47	Shirley Chisholm State Park	56	0.29	No	No	No	Yes	NA	No
87	Turtle Cove	57	0.29	No	No	Yes	Yes	No	Yes
57	Little Brother Island South	58	0.29	No	No	Yes	Yes	Yes	Yes
15	Fort Totten	59	0.28	No	No	Yes	NA	NA	Yes
14	Fort Totten	59	0.28	No	No	Yes	NA	NA	Yes

***Confidence Layer of Water Quality Score:**

High
 Moderate +
 Moderate
 Moderate -
 Low

Table 15. Summary of Site Rankings and Associated Environmental and Physical Conditions (Ranks 61-78).

ID	Site	Water Quality Parameters		Physical Parameters			Built Environment		
		Rank	Score	Wave Exposure	Erosion	Depth	Near CSO	Near MS4	Near Park
43	Little Bay Park	61	0.28	No	No	Yes	NA	NA	Yes
88	City Island	61	0.28	No	No	Yes	Yes	No	Yes
62	Ivington - South	63	0.28	No	No	No	NA	NA	Yes
48	Shirley Chisholm State Park	64	0.27	No	No	Yes	Yes	NA	No
1	Piermont - South	65	0.27	No	No	No	NA	NA	Yes
86	Pugsley Creek	66	0.26	No	No	Yes	NA	NA	Yes
0	Piermont - North	67	0.25	No	No	No	NA	NA	Yes
63	Irrington - North	67	0.25	No	No	No	NA	NA	Yes
89	City Island	69	0.24	No	No	Yes	NA	NA	Yes
85	Powell's Cove	69	0.24	No	No	Yes	Yes	No	Yes
20	Malba	69	0.24	No	No	Yes	Yes	No	Yes
32	Bush Terminal - USACE	72	0.22	No	No	No	NA	NA	No
33	Bush Terminal - USACE	72	0.22	No	No	No	NA	NA	Yes
35	Bush Terminal - USACE	74	0.22	Yes	No	No	NA	NA	No
34	Bush Terminal - USACE	74	0.22	Yes	No	No	NA	NA	Yes
72	Bush Terminal Park	74	0.22	No	No	No	NA	NA	Yes
99	Ossining	77	0.2	No	No	No	NA	NA	Yes
100	Earle	77	0.2	No	No	No	NA	NA	Yes

***Confidence Layer of Water Quality Score:**

High
 Moderate +
 Moderate
 Moderate -
 Low

7.2. Limitations of the Study

The results presented in this study are subject to several limitations related to data availability, quality, and consistency. All analyses rely on existing datasets, which inherently contain varying levels of uncertainty that are not fully captured within the framework. These uncertainties arise from differences in data collection methods, instrumentation, and reporting standards across sources.

Data sparsity, both in sampling frequency and spatial coverage, represents a key limitation, as monitoring is uneven across locations and time. Some sites are supported by dense and recent observations, while others rely on limited or infrequent measurements, which may affect the robustness of site-level estimates. In addition, heterogeneity in sampling design, including differences in sampling frequency, depth, seasonal coverage, and measured parameters, introduces further variability that can influence comparability across sites.

To mitigate these limitations, efforts were made to harmonize datasets as much as possible. This included standardizing units, applying consistent filtering criteria, aligning temporal windows, and implementing uniform analytical methods across all data sources. In parallel, uncertainty was explicitly evaluated and reported using statistical approaches (e.g., bootstrap resampling) and spatial data support metrics. These steps were intended to reduce inconsistencies and provide a transparent assessment of confidence in the results.

Despite these efforts, residual uncertainty remains, particularly in areas with limited data coverage. Consequently, the results should be interpreted as the best possible representation given current data availability and methodological constraints. The rankings and associated confidence levels reflect existing knowledge rather than absolute conditions, and may evolve as additional data become available or as monitoring efforts improve.

Appendices

Appendix A: Natrx Assess - Shoreline Change Analysis (SCA)

Marsh Edge from Image Processing (MEIP) methodology applied to National Aerial Imagery Program (NAIP) imagery (2010-present) using area-based analysis (ABA) techniques to quantify historical coastal change patterns and rates. MEIP methodology utilizes the vegetation line as a proxy for shoreline position. The Assess platform incorporates human-in-the-loop quality control and validation processes to ensure data accuracy and reliability.

Analysis outputs include:

- **land_change_ft_per_year**: Rate of shoreline erosion or accretion in feet per year, derived from linear regression through all temporal data points to extract dominant trend signals
- **epr_ft_per_year**: End Point Rate representing linear change between initial and final measurements divided by analysis period
- **r_squared**: Statistical confidence measure indicating proportion of variance explained by the linear regression model
- **nsm_ft**: Net Shoreline Movement representing total positional change between initial and final measurements

Appendix B: Natrx Assess - Wind-driven Wave Modeling

Computational analysis using USACE wind fetch model equations from the Shore Protection Manual (USACE 1984) and Coastal Engineering Manual (USACE 2007) to determine wave characteristics essential for oyster reef design. The analysis incorporates fetch distance calculations, friction velocity coefficients, and drag coefficients to model wave energy distribution. Output parameters include significant wave height, spectral peak wave period, directional wave distributions, and exceedance probabilities. These datasets inform optimal reef structure placement, design wave criteria, and site-specific conditions affecting both oyster survival and structure stability.

Appendix C: Robustness of Site Rankings Across Water Quality Aggregation Methods

The heatmap compares site ranks across three aggregation approaches, with rows ordered by the DO-modifier method. Consistent color patterns indicate stable rankings, while shifts highlight sensitivity to aggregation. Alternative methods were used to test the robustness of the DO-modifier ranking, ensuring that it reflects the underlying data rather than the aggregation approach.

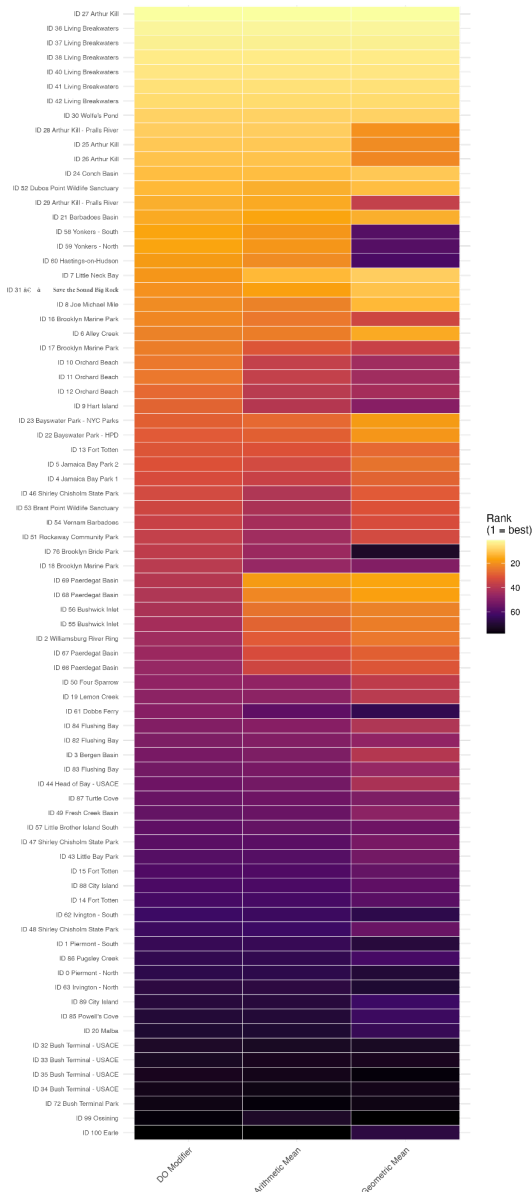


Figure C1. Rank Robustness Across Aggregation Methods.

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